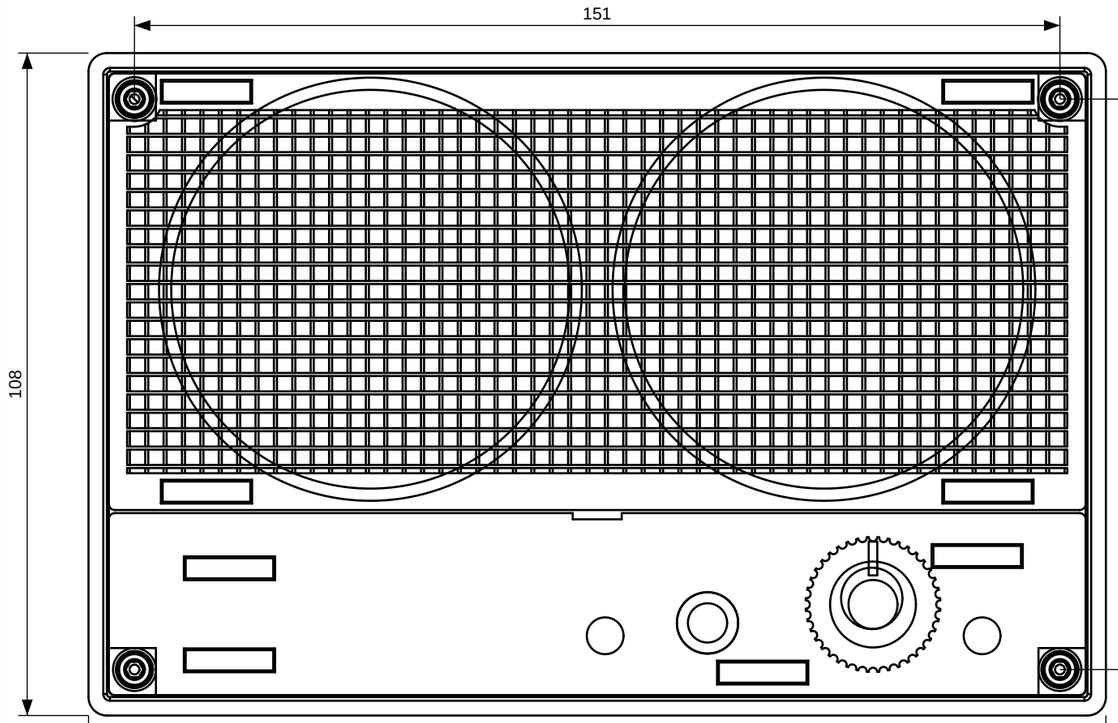
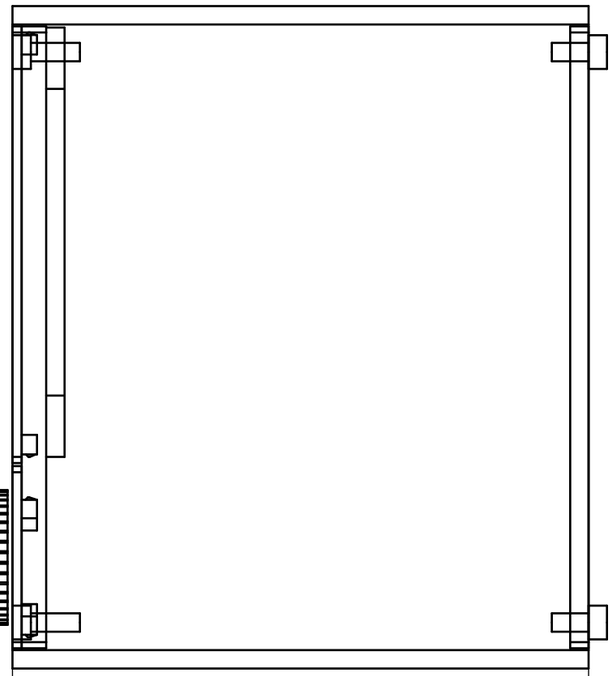
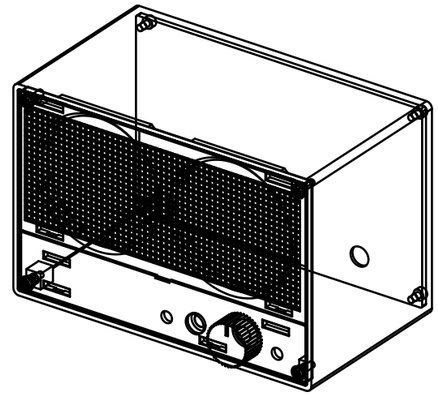
FRONT



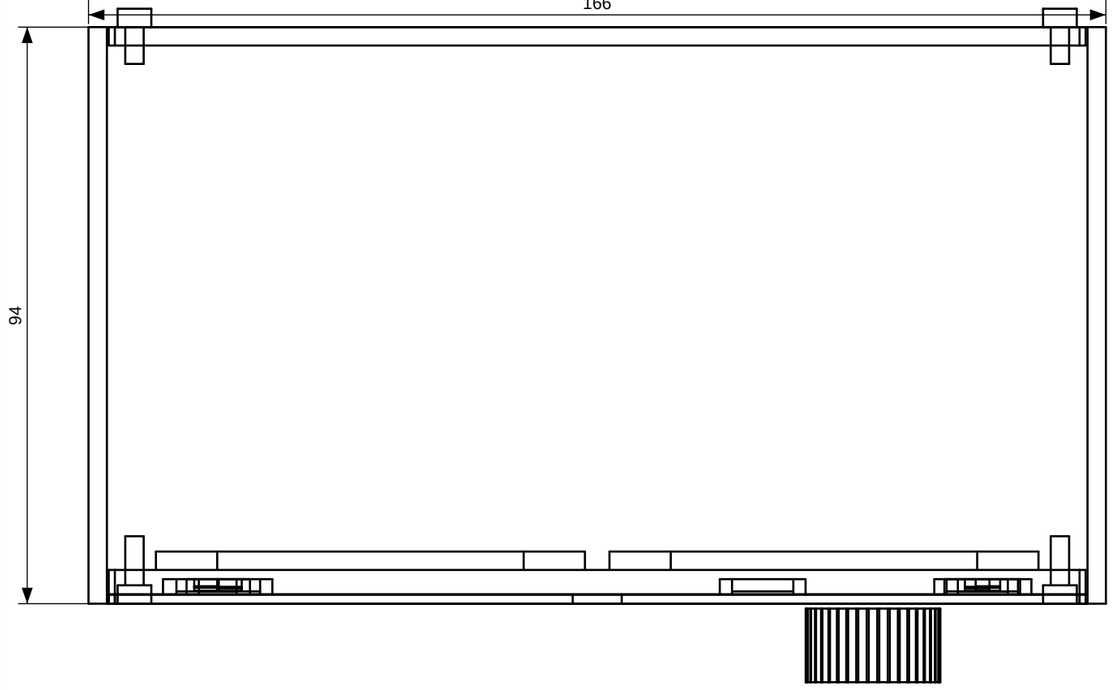
RIGHT



ISOMETRIC (0.36:1)



TOP

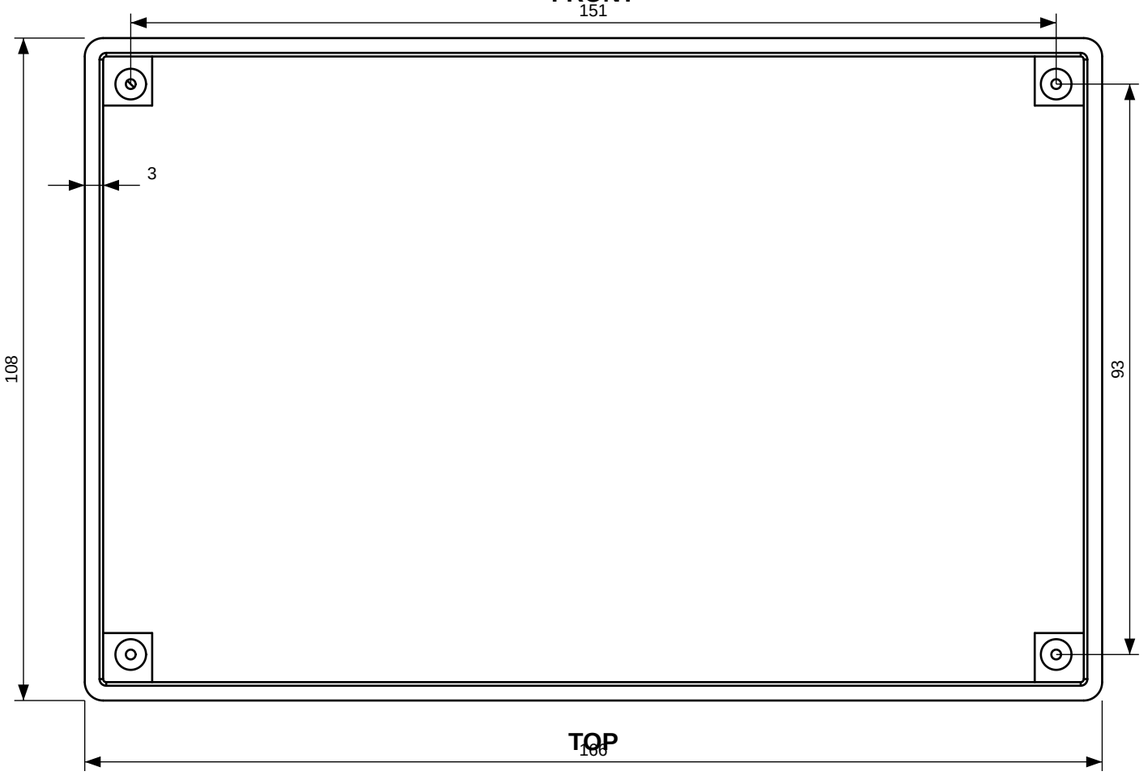


Grille (grey) and control strip (white) snap onto the orange baffle. Four M3 socket head cap screws show in the corners, flush with the cover faces. Feet, drivers, and the amplifier are not shown.

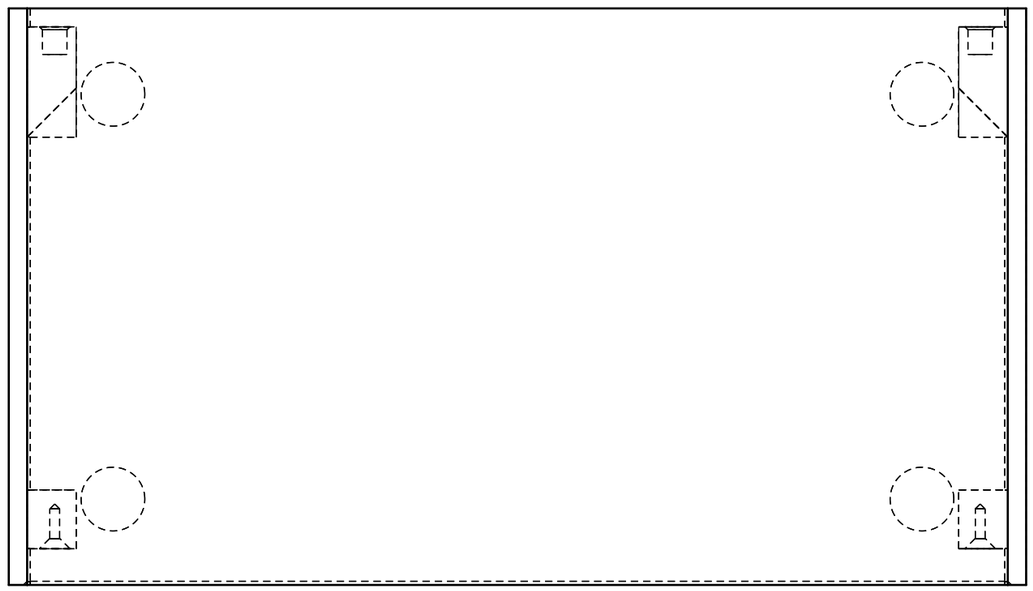
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: General arrangement				
Drawing no.: BSM-00	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 1 / 12
Material: PLA (printed parts)			Tolerance unless shown · units: ±0.2 · mm · third angle	

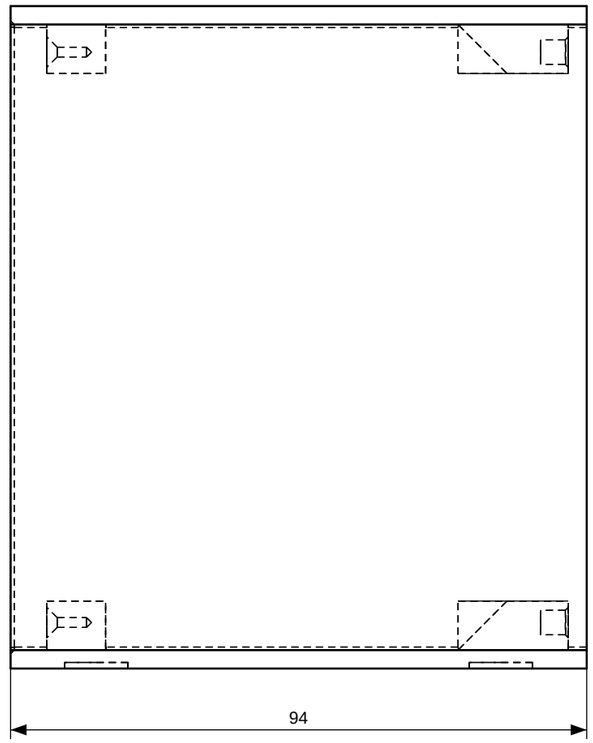
FRONT



TOP



RIGHT

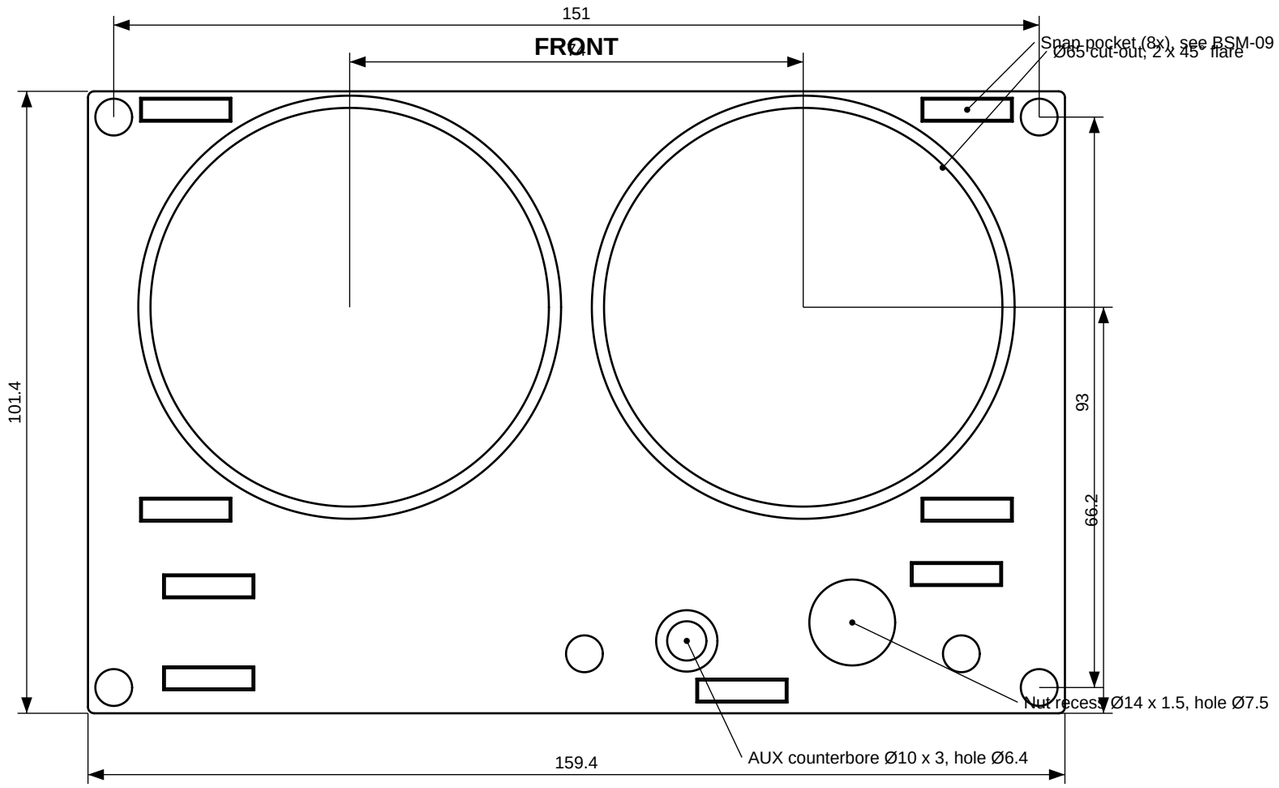


Print front opening down. Profile: PLA Shell 0.25.
Supports from the build plate below the four front posts only.
Front posts: pilot Ø1.6, drill Ø4.0 x 7 after the print, insert M3 x 3.
Rear posts: printed Ø4.0, 0.5 x 45° relief, 45° gussets.
Lead-in chamfer 0.6 x 45° inside the front opening.
Panel opening 160.0 x 102.0 (panels 159.4 x 101.4).

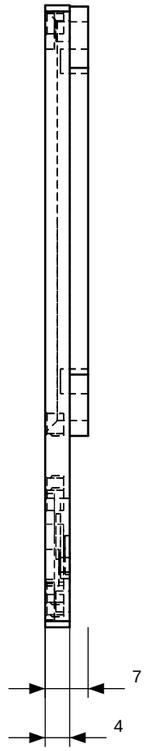
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Sleeve				
Drawing no.: BSM-01	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 2 / 12
Material: PLA, grey			Tolerance unless shown · units: ±0.2 · mm · third angle	

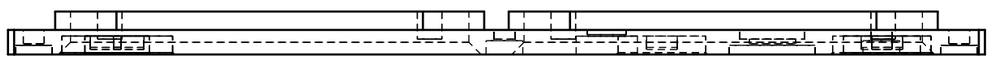
FRONT



RIGHT



TOP

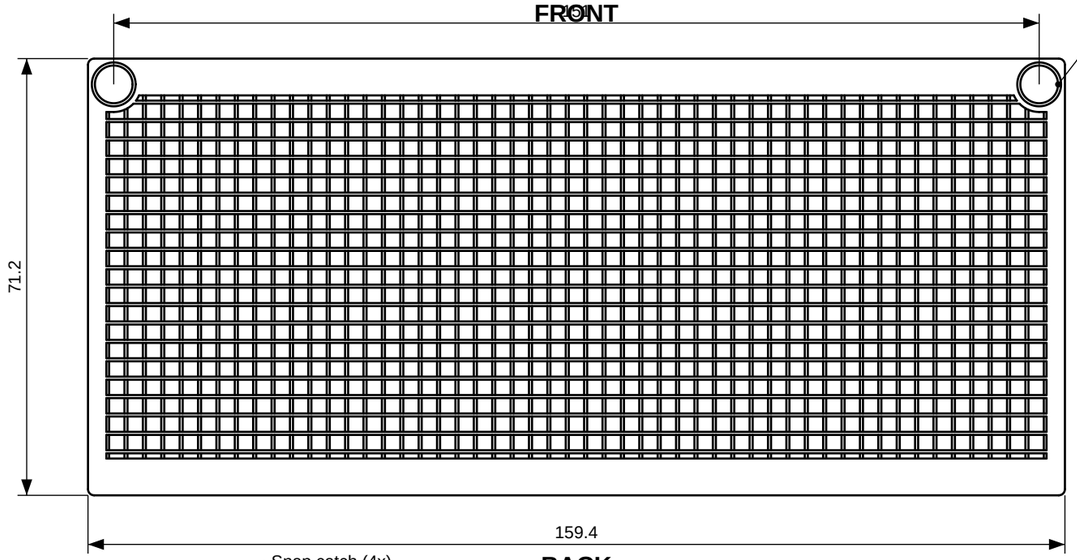


Print front face down. Profile: PLA Parts 0.25. No supports.
Counterbores Ø6.0 x 1.5 (front) and x 2.25 (tray), membrane 0.25: drill Ø3.4 from the back after the print.
Cut the pot and AUX membranes with a knife.
Spacer frames 70 x 70, 3 thick, 10 x 45° corners: 8 inserts M3 x 3.
Snap pockets: 3.0 deep, 45° lip, see BSM-09.

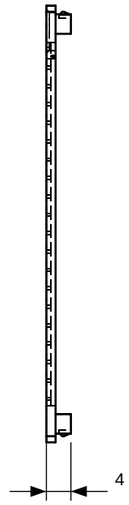
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Front baffle				
Drawing no.: BSM-02	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 3 / 12
Material: PLA, orange (shows through the grille)			Tolerance unless shown · units: ±0.2 · mm · third angle	

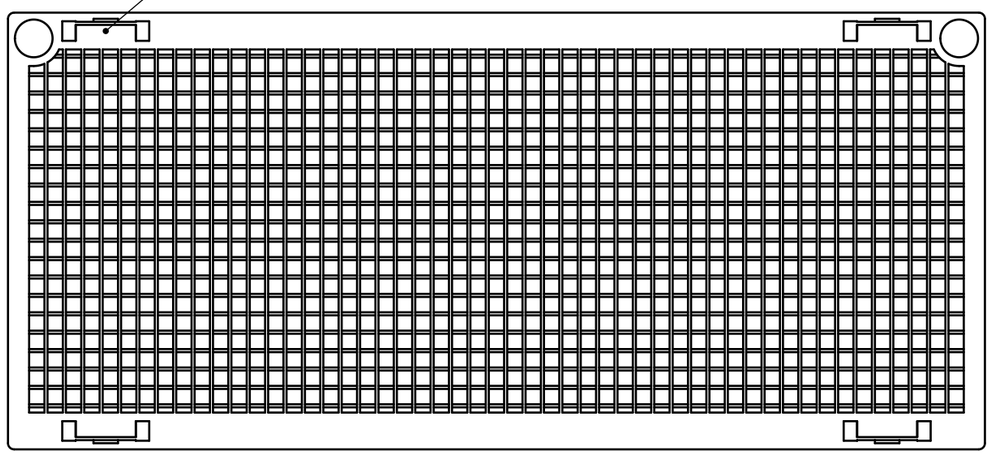
FRONT



RIGHT



BACK

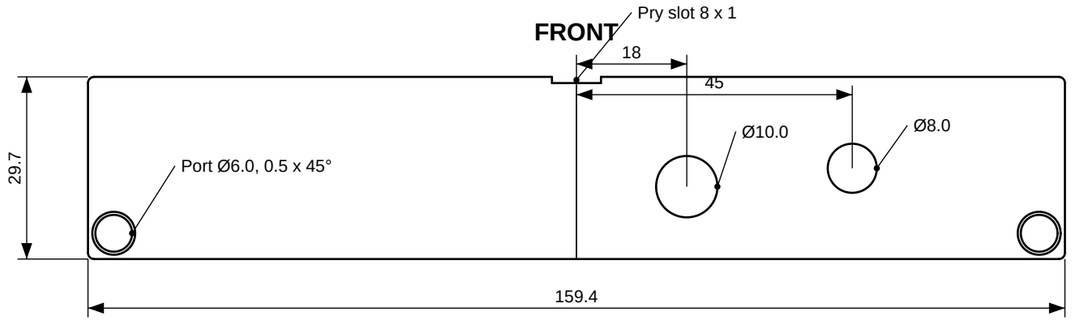


Print front face down. Profile: PLA Grille 0.25 (no expansion).
Ribs: one 0.5 line, 3.0 pitch, horizontal on layers 1-3,
vertical on layers 4-6. About 69 % open.
Ports and catches carry the expansion in the model:
ports +0.075 per side, catch posts and bumps +0.095.
Snap catches: see BSM-09.

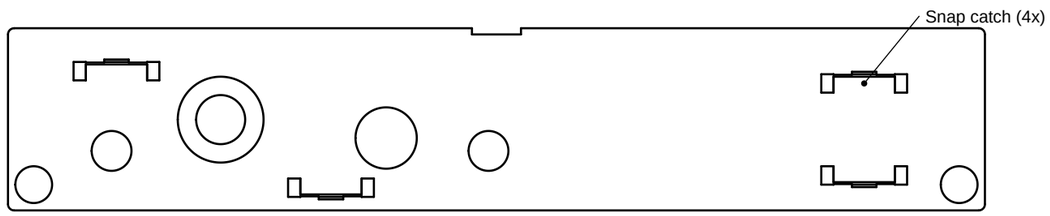
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Front grille (lattice)				
Drawing no.: BSM-03	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 4 / 12
Material: PLA, grey			Tolerance unless shown · units: ±0.2 · mm · third angle	

FRONT



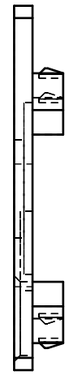
BACK



TOP



RIGHT (2:1)

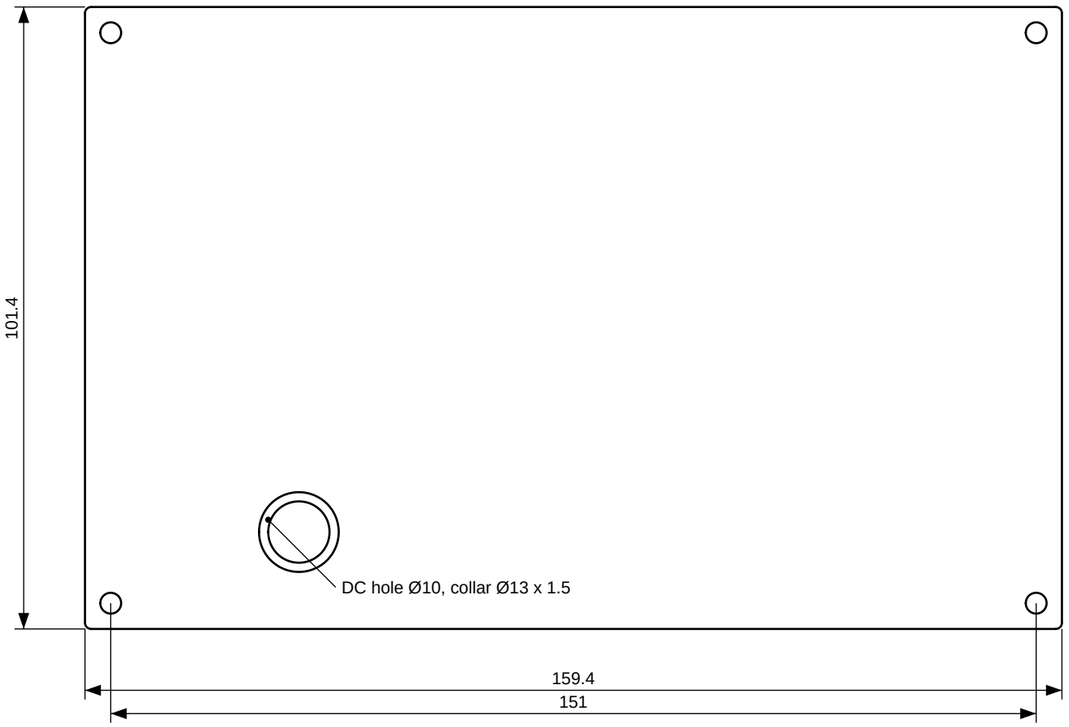


Print front face down. Profile: PLA Parts 0.25. Colour white.
Snap beam modelled 0.31: with 0.095 expansion it prints one 0.5 line.
Back: two pockets Ø6.5 x 1.0 for the tray screw heads.
Snap catches: see BSM-09.

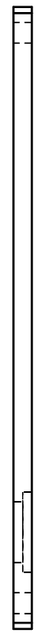
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Control strip				
Drawing no.: BSM-04	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 5 / 12
Material: PLA, white			Tolerance unless shown · units: ±0.2 · mm · third angle	

BACK



RIGHT

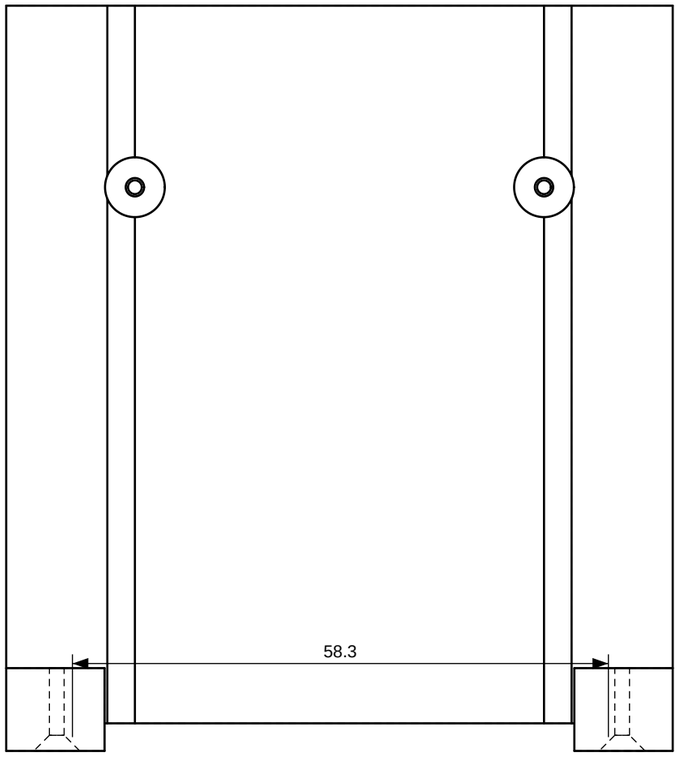


Print outer face down. Profile: PLA Parts 0.25.
Four Ø3.4 clearance holes for M3 x 6 socket head cap screws.
No counterbore: the heads sit on the panel.

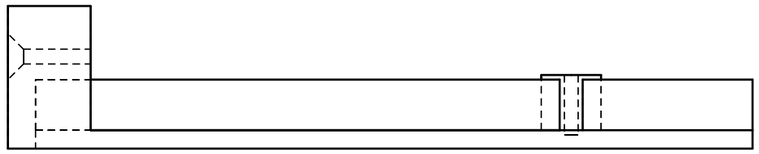
Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Rear panel				
Drawing no.: BSM-05	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 6 / 12
Material: PLA, grey			Tolerance unless shown · units: ±0.2 · mm · third angle	

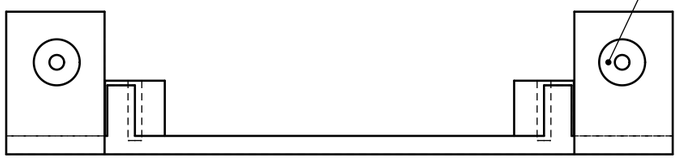
TOP (1.5:1)



RIGHT (1.5:1)



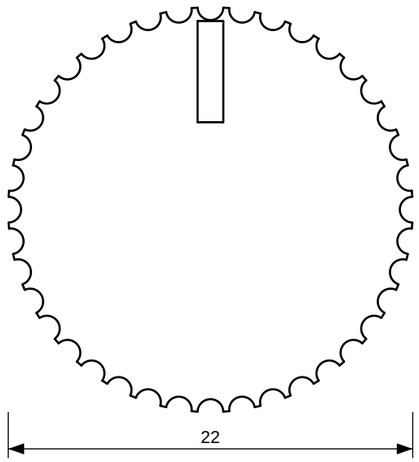
FRONT (1.5:1)



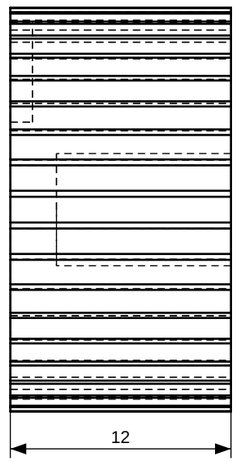
Print flat. Profile: PLA Parts 0.25.
Rails carry the ZK-502C board 8.0 above the sleeve floor.
The two tabs screw to the baffle back (M3 x 8).

Title: **Bluespeaker Mini**				
Subject: Amplifier tray				
Drawing no.: BSM-06	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 7 / 12
Material: PLA, grey			Tolerance unless shown · units: ±0.2 · mm · third angle	

FRONT (3:1)



RIGHT (3:1)

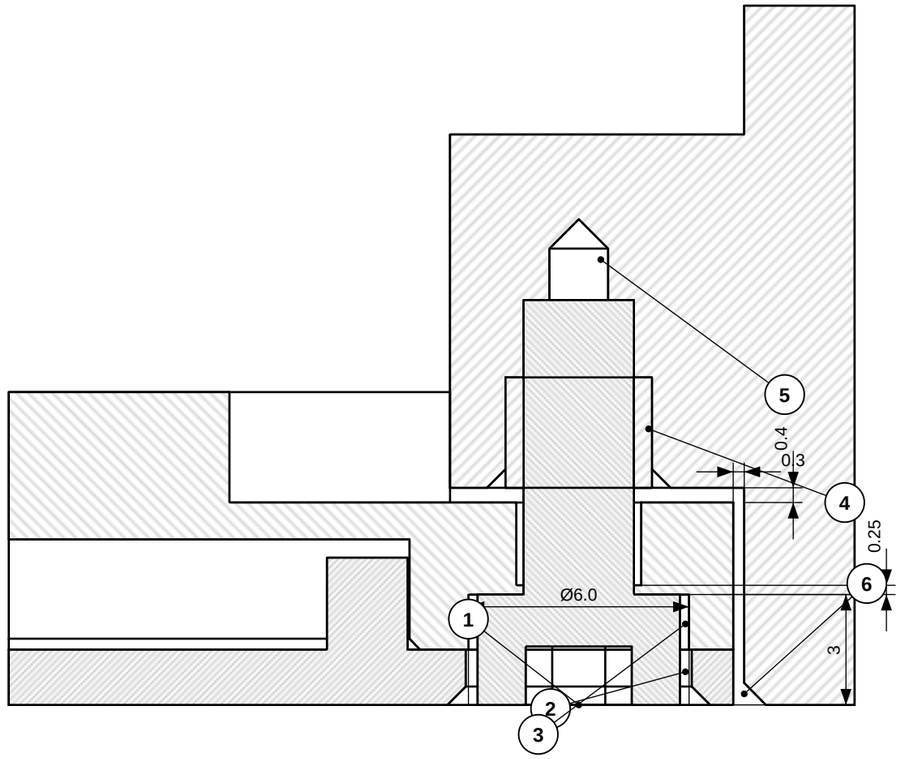


Print front face up. Profile: PLA Parts 0.25. Colour orange.
Bore Ø6.1 x 9.5 for the volume control shaft (check the shaft type).
Glue the white knob mark (1.4 x 1.2 x 5.5) into the groove.

Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

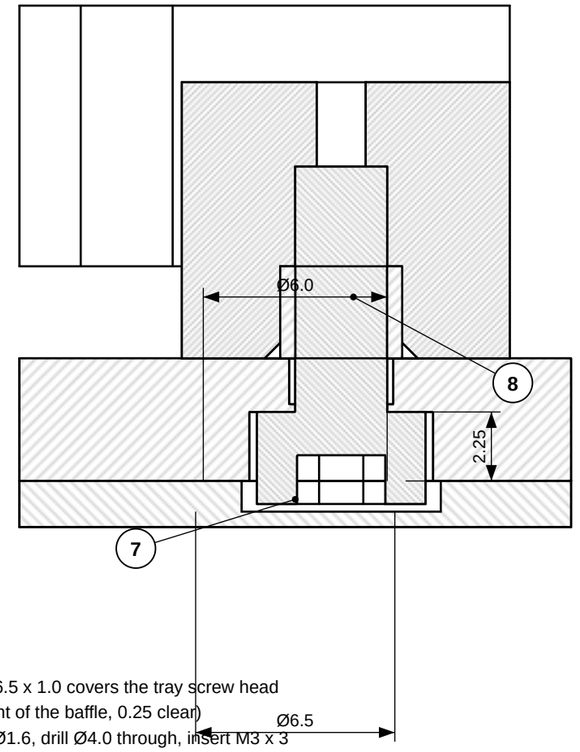
Title: **Bluespeaker Mini**				
Subject: Knob				
Drawing no.: BSM-07	Revision: 0.3	Date: 2026-10-11	Scale: 1:1	Sheet: 8 / 12
Material: PLA, orange			Tolerance unless shown · units: ±0.2 · mm · third angle	

SECTION A-A (Z = 100.5, seen from above) 6:1



1 M3 x 8 socket head cap screw, head flush with the grille face
2 Grille port Ø6.15 (Ø6.0 + 0.075 per side), 0.5 x 45° chamfer
3 Baffle counterbore Ø6.0 x 1.5, membrane 0.25: drill Ø3.4
4 Insert M3 x 3 (short), in the sleeve front post
5 Pilot Ø1.6 with 45° tip: drill Ø4.0 x 7
6 Sleeve lead-in chamfer 0.6 x 45°; panel fit gap 0.3

SECTION E-E (Z = 13.0) 5:1



7 Strip pocket Ø6.5 x 1.0 covers the tray screw head
(head 0.75 in front of the baffle, 0.25 clear)
8 Tray tab: pilot Ø1.6, drill Ø4.0 through, insert M3 x 3
Counterbore Ø6.0 x 2.25, membrane 0.25

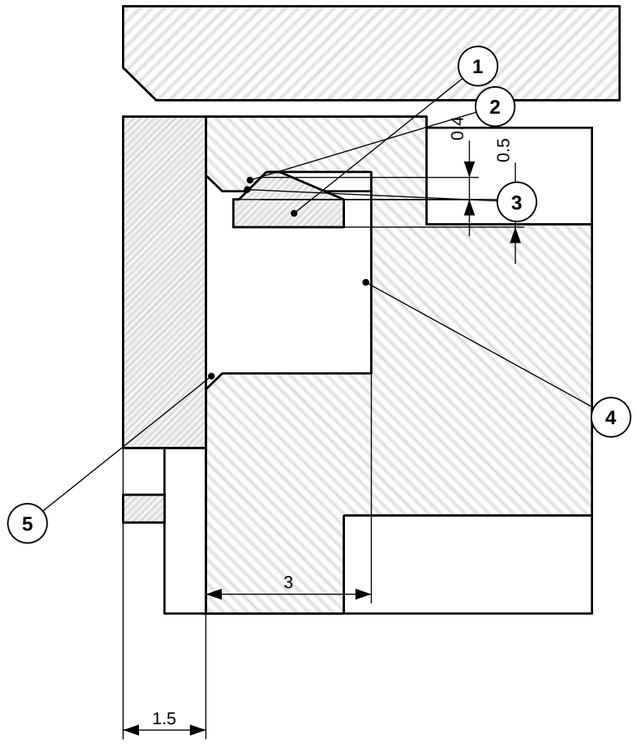
Title
Bluespeaker Mini

Subject
Joint details: front corner and tray screws

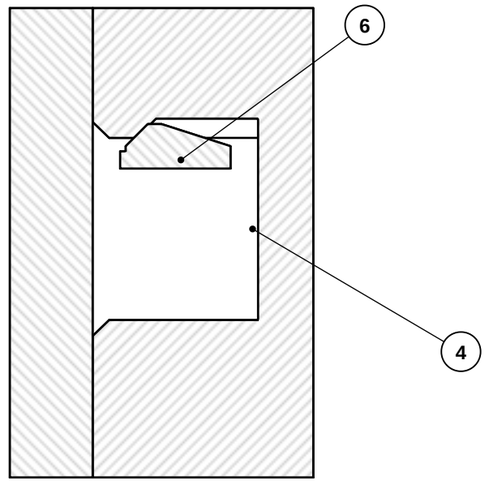
Drawing no.	Revision	Date	Scale	Sheet
BSM-08	0.3	2026-10-11	6:1, 5:1	9 / 12

Material	Tolerance unless shown · units
PLA	±0.2 · mm · third angle

SECTION B-B grille catch (X = 63.75) 9:1



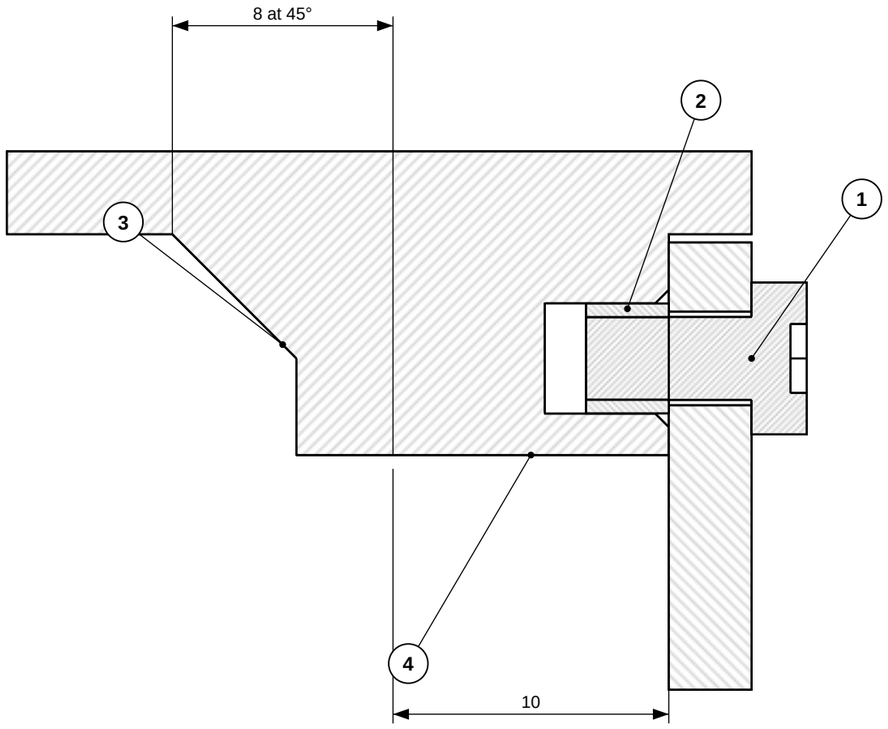
SECTION F-F strip catch (X = 62.0) 9:1



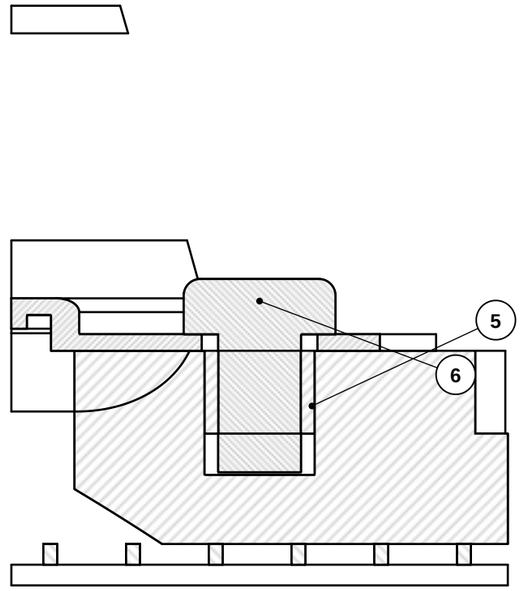
1 Beam: one 0.5 line, 10 long, 2.0 high; bridges 0.5 above the cover back. Flexes in the cover plane.
2 Bump 0.4 (grille +0.095 comp): 45° catch face, flat 0.25, 18° lead-in
3 Lip at the pocket mouth: 45°, 0.05 preload on the bump (pulls the cover flat)
4 Pocket 3.0 deep, 3.3 wide, floor is a bridge. Posts 0.15 clear per side (locate the cover)
5 Mouth lead-in 0.3 x 45° (takes the elephant foot)
6 Strip beam modelled 0.31: the Parts profile expansion (0.095) prints one 0.5 line

Title: **Bluespeaker Mini**				
Subject: Joint details: snap catches				
Drawing no.: BSM-09	Revision: 0.3	Date: 2026-10-11	Scale: 9:1	Sheet: 10 / 12
Material: PLA			Tolerance unless shown · units: ±0.2 · mm · third angle	

SECTION G-G rear post (X = 75.5) 4.5:1



SECTION H-H driver mount (Z = 95.59) 4.5:1



1 M3 x 6 socket head cap screw; head on the rear panel, Ø3.4 clearance
2 Insert M3 x 3, printed Ø4.0 hole, 0.5 x 45° relief chamfer
3 Gussets at 45° to both walls: the post prints with no support
4 Post 8 x 8, 10 deep
5 Insert M3 x 3 in the 3 mm spacer frame
6 M3 x 5 pan head through the driver flange

Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

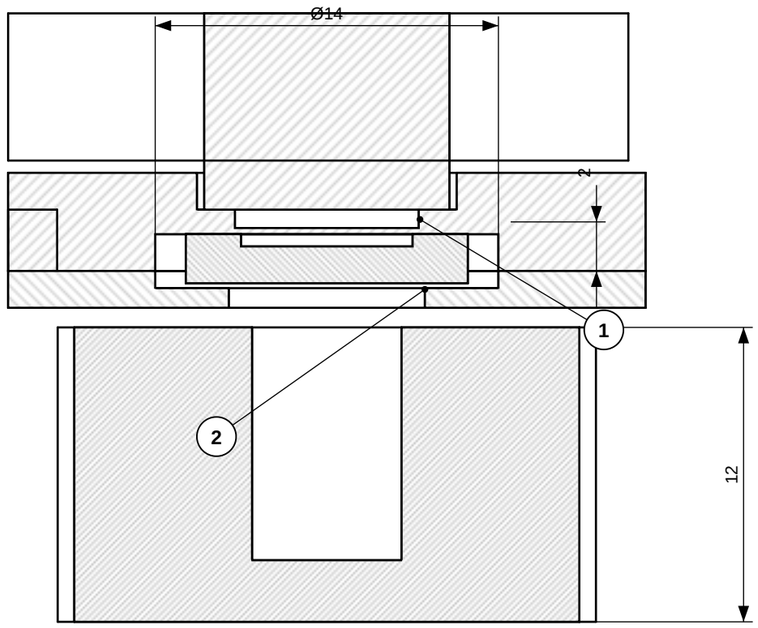
Title
Bluespeaker Mini

Subject
Joint details: rear post and driver mount

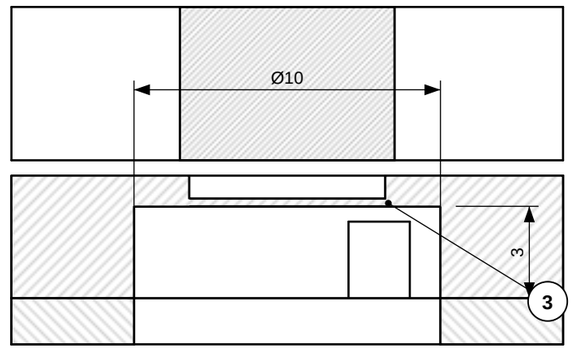
Drawing no.	Revision	Date	Scale	Sheet
BSM-10	0.3	2026-10-11	4.5:1	11 / 12

Material	Tolerance unless shown · units
PLA	±0.2 · mm · third angle

SECTION J-J volume control (Z = 18.1) 4:1



SECTION K-K AUX socket (Z = 15.1) 5:1



1 Membrane 0.25 at the floor of the Ø14 x 1.5 nut recess: cut out to Ø7.5
2 Strip hole Ø8.0; knob Ø22 x 12 with 0.8 gap to the strip
3 Membrane 0.25 at the floor of the Ø10 x 3 collar counterbore: cut out to Ø6.4
The ZK-502C positions come from photos. Measure the board before production.

Bluespeaker Mini · build123d model cad/bluespeaker_mini.py

Title: **Bluespeaker Mini**				
Subject: Joint details: volume control and AUX socket				
Drawing no.: BSM-11	Revision: 0.3	Date: 2026-10-11	Scale: 4:1, 5:1	Sheet: 12 / 12
Material: PLA			Tolerance unless shown · units: ±0.2 · mm · third angle	